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ECPE 136
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Lab Assignment 7 - 3ip Nand & 3ip NOR Layout

Objective:

- The objective of this lab is to learn how to make a layout using cadence virtuoso. Within this lab we will reuse the schematics we made in lab 5 of three input NAND and NOR gates. This lab will use those labs to initialize our layout with the same inputs and outputs. We will then use the software to create our layers of metal, poly, and vias to create the inputs and output for the two types of design schematics. The NAND logic will have three pmos in parallel and three nmos in series. The NOR layout will have three pmos in series and three nmos in parallel.

Procedures:

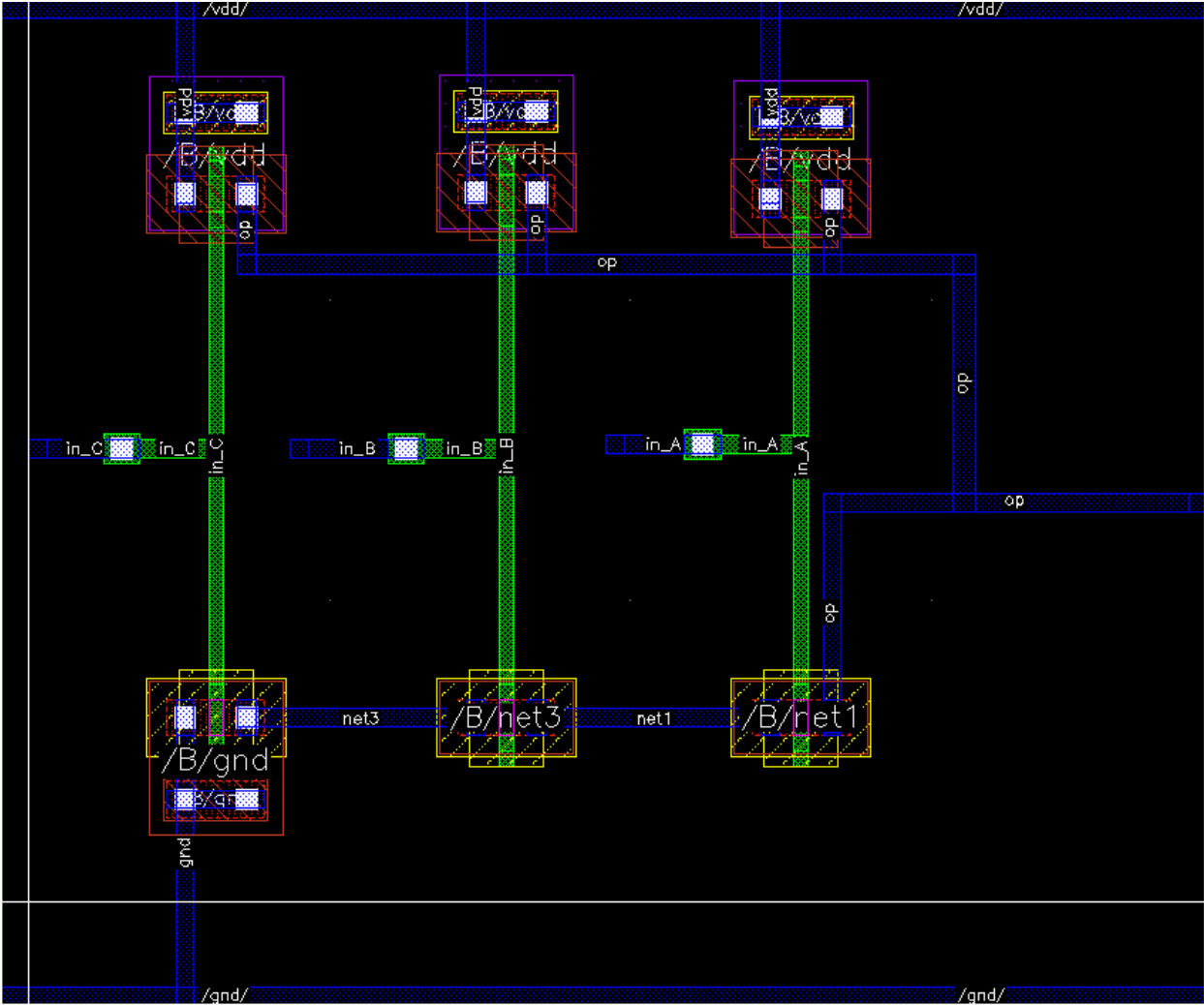
1.0 - Create a Layout

1. Using the previous schematics of NOR and NAND gates we will choose the 'Layout XL'. This will create a layout screen pre populated with the nmos and pmos gates in the same layout as your schematics.
2. For NAND layout, we need to setup the pmos detached so that each connection to vdd can be made for each three pmos and then the output will be derived to the output variable. The nmos in series need to connect one to ground and then the next get connected to each other to form the series connections with the final nmos carrying the output to the output pin.
3. For NOR layout, we need to setup the pmos detached so that each connection to gnd can be made for each three pmos and then the output will be derived to the output variable. The nmos in series need to connect one to ground and then the next get connected to each other to form the parallel connections with the final nmos carrying the output to the output pin.

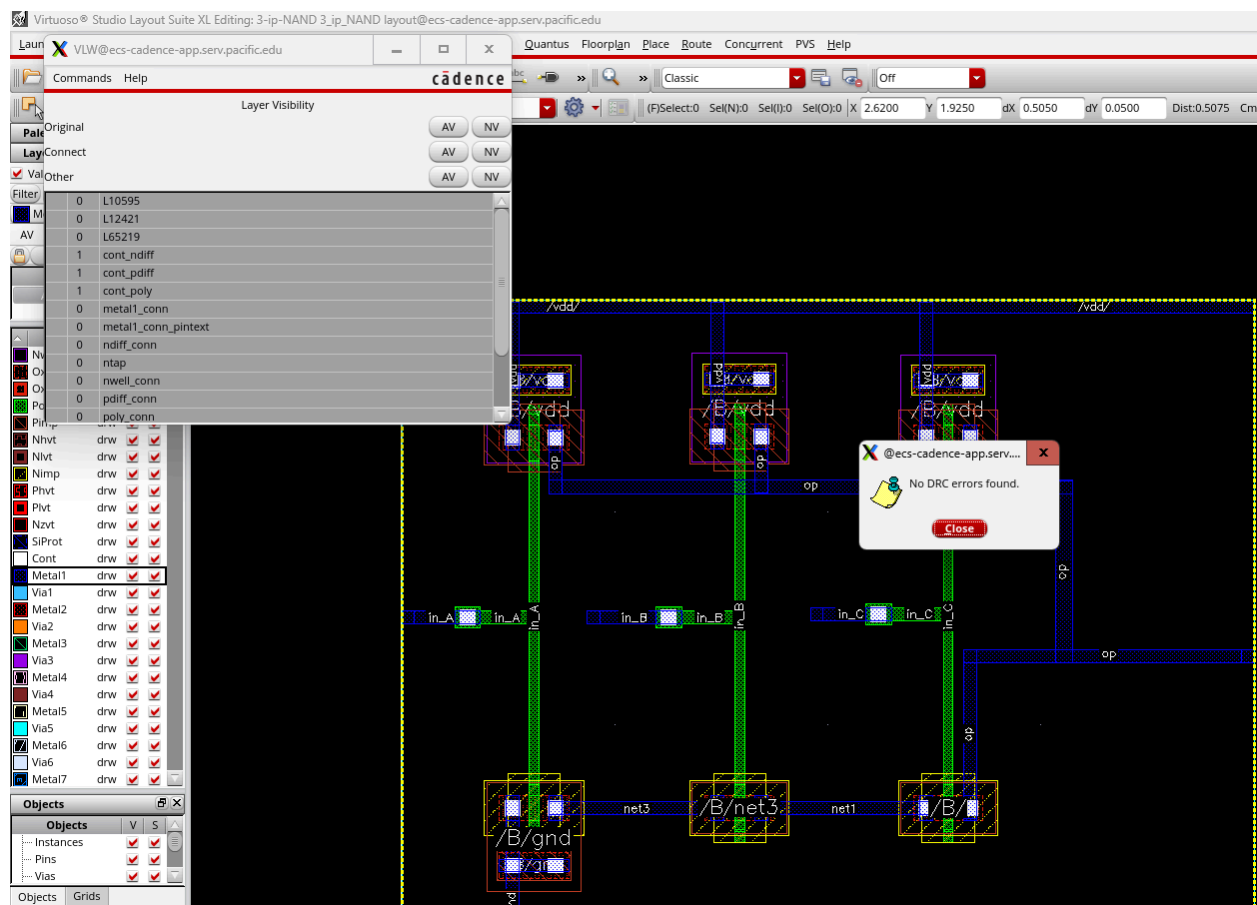
2.0 - Create a Waveform

1. To confirm that the layouts are correctly made we need to run a DRC check to verify that the layout does not have any errors. When this completes and detects no errors we need to run a LVS check before creating a waveform. When this is done checking and no errors we can open the maestro tab from the previous lab and input the simulation profile file created from the LVS check. This then can run and will generate a waveform to verify the input and outputs.

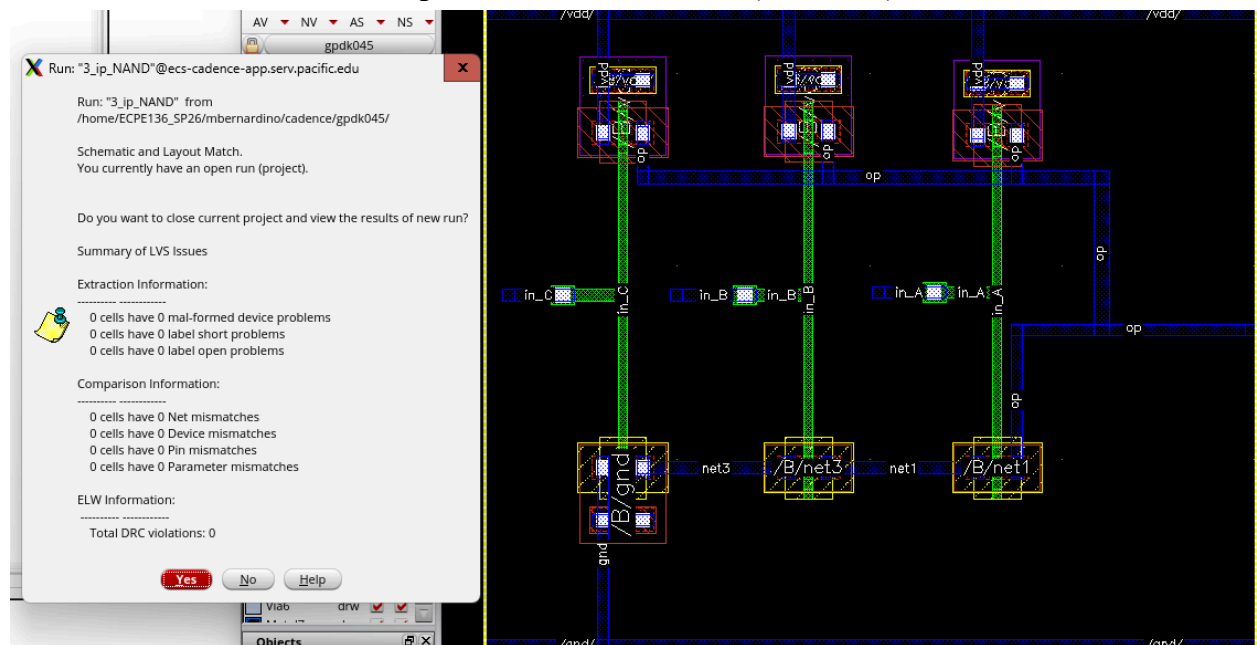
Design description:



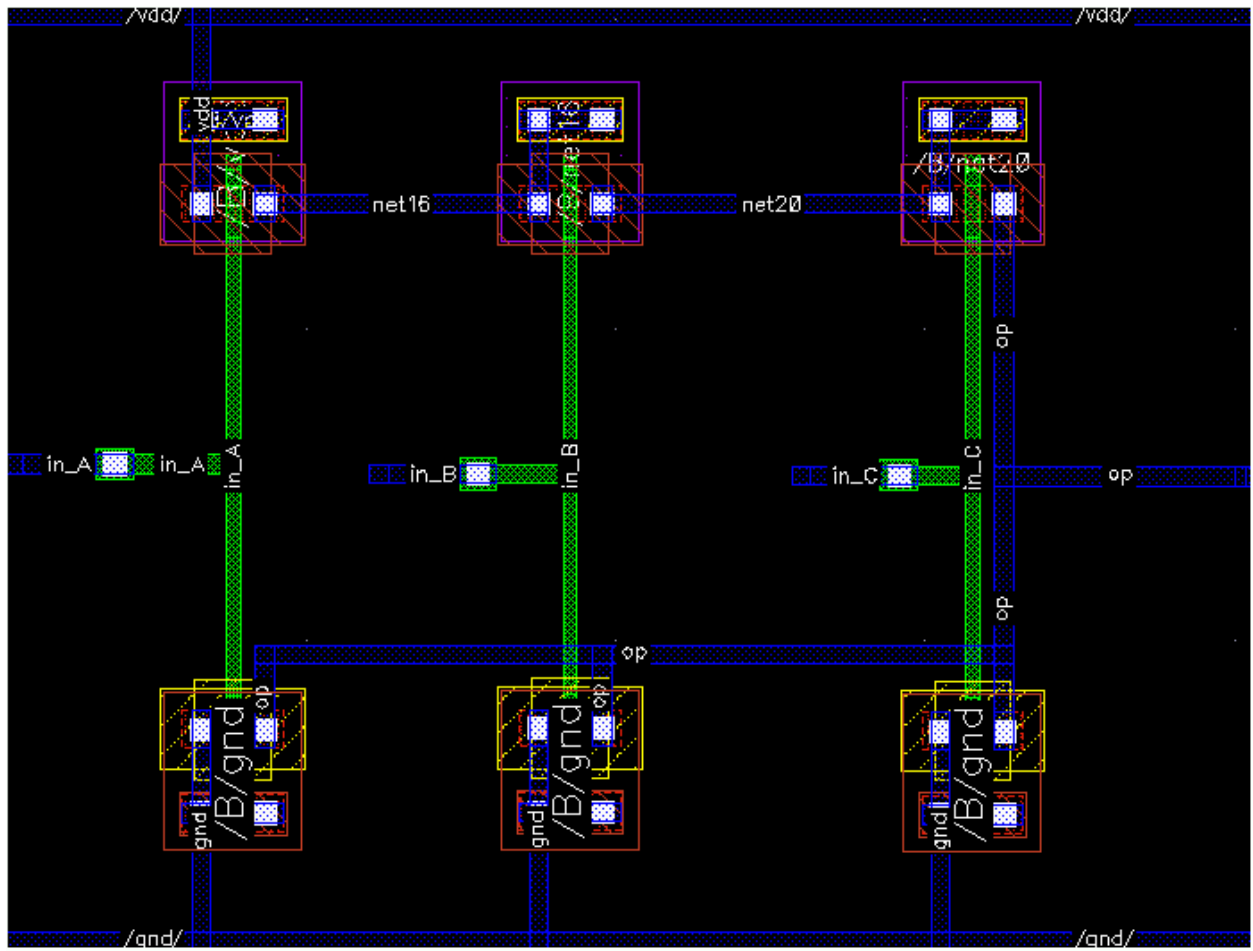
3-Input NAND Layout



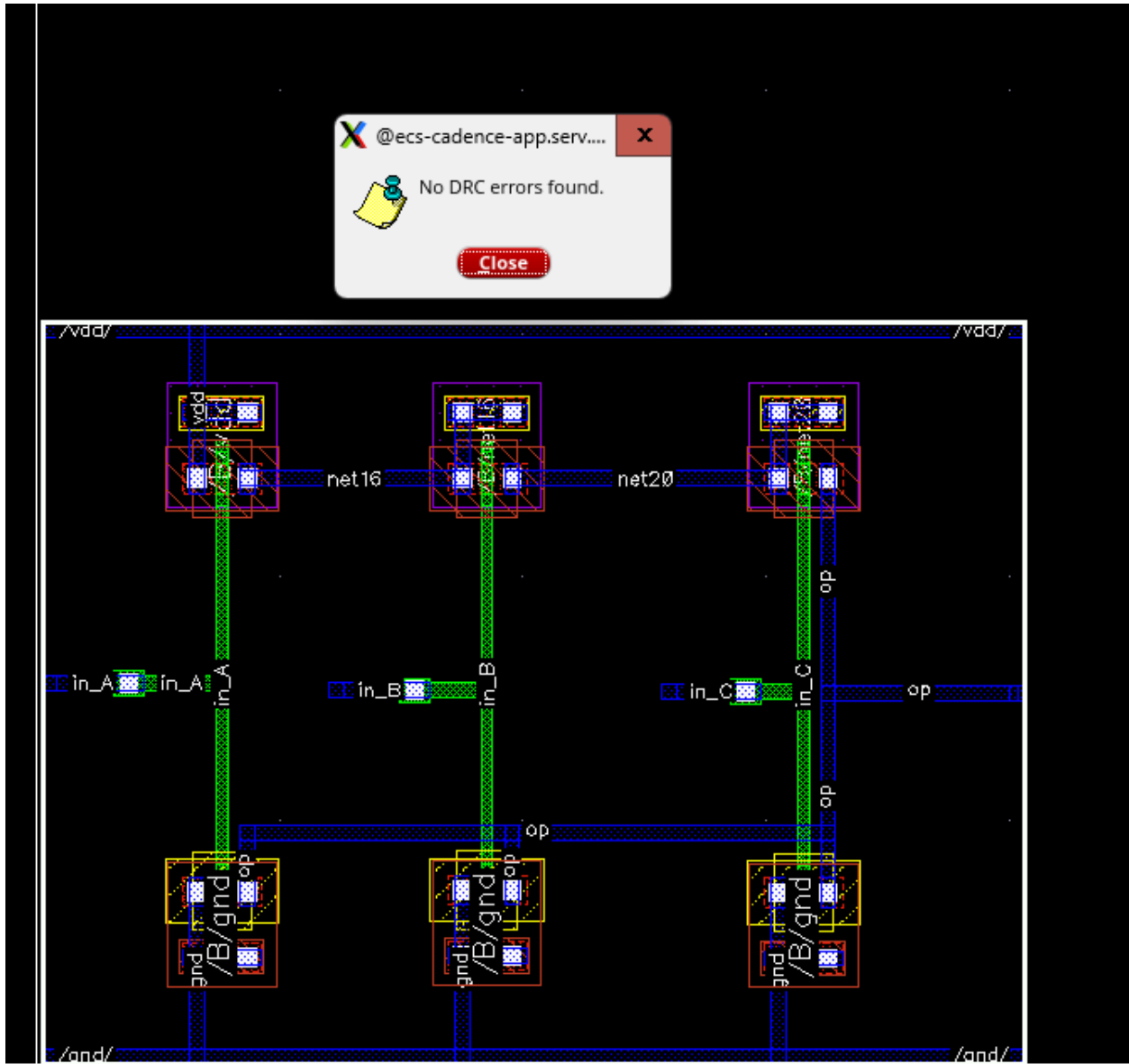
3-Input NAND DRC results (no errors)



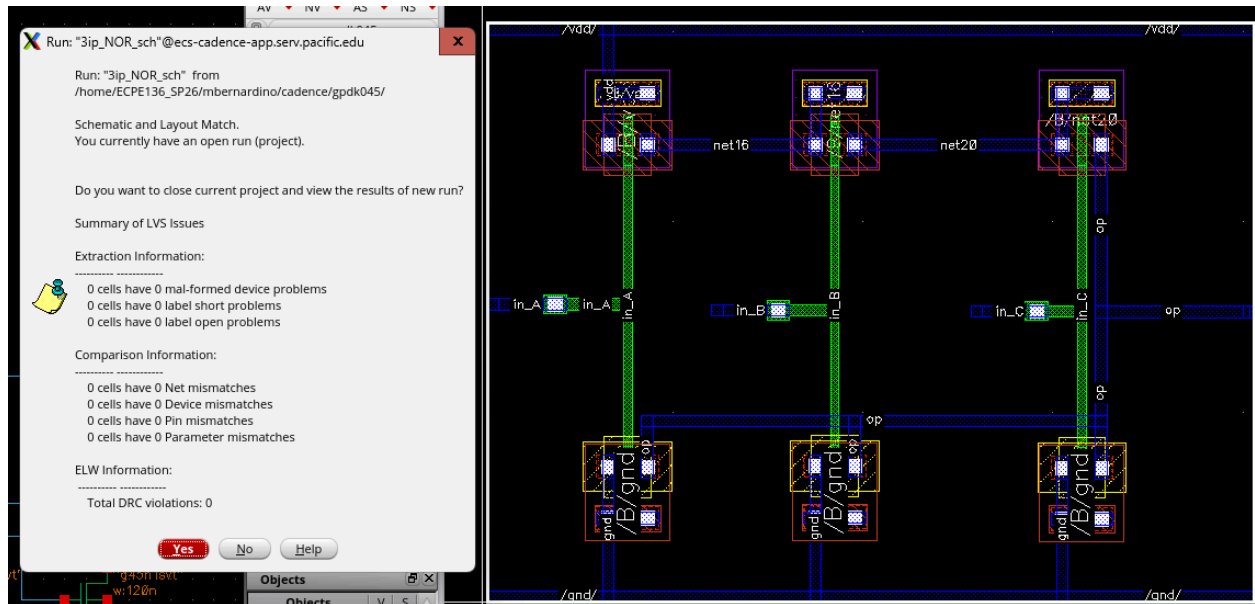
3-Input NAND LVS results (no error)



3-Input NOR Layout

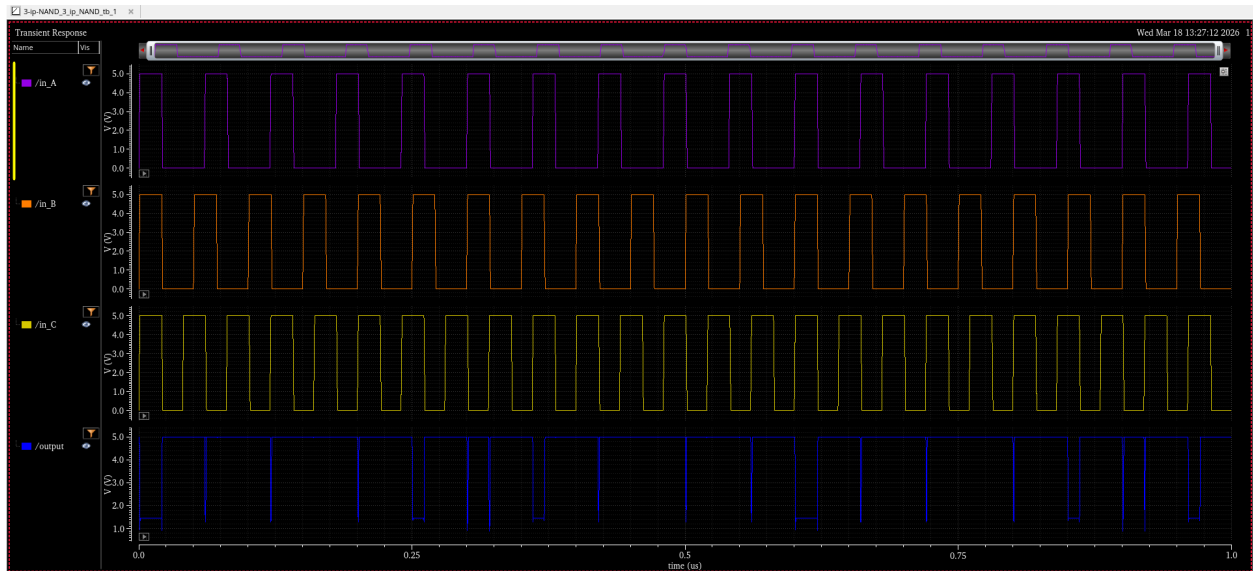


3-Input NOR DRC results



3-Input NOR LVS results

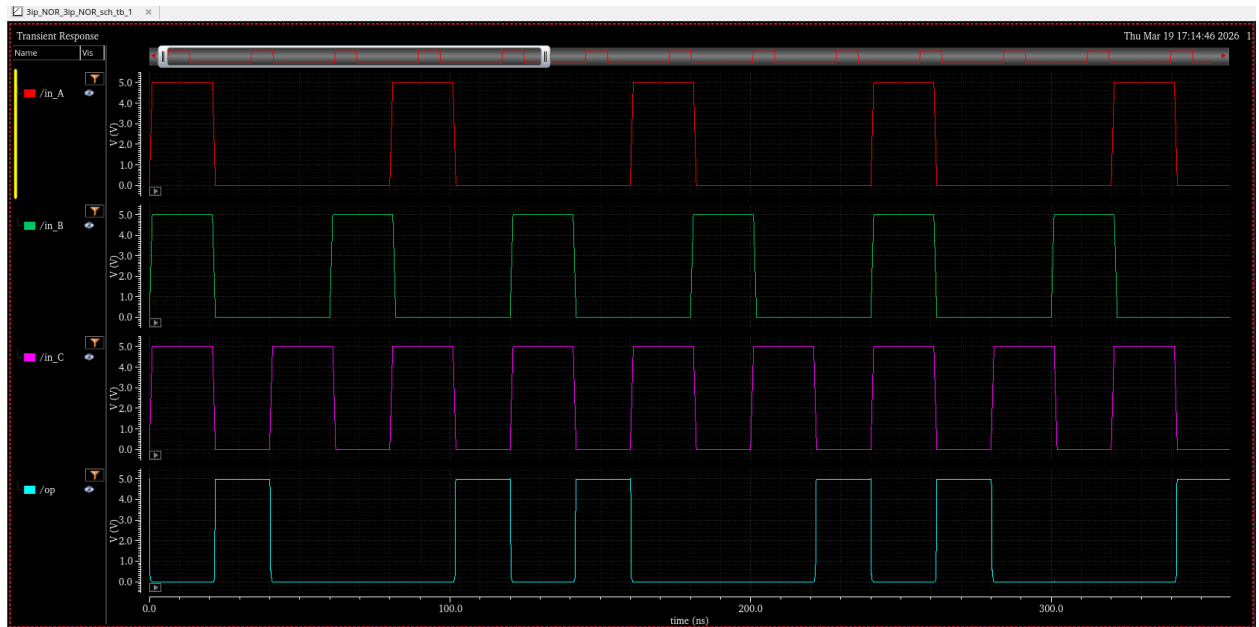
Results & Observations:



3-Input NAND Waveform

Observation:

While I was unable to create a waveform due to Quantus (Assura) generation errors. I have the waveform of what it should look like from the previous lab. We should be seeing the high (1) when any pmos is active since they are in parallel. When we see low (0) then the nmos are all on as they are in series.



3-Input NOR Waveform

Observation:

- When comparing my waveform from my NOR generated from completing the LVS stage. The waveform is correct and accurate. In a NOR the pmos are in series and the nmos are in parallel. When the nmos is on (1) then a path to gnd is shown at output. When all pmos are on (0) then a path to vdd is active.

Conclusion:

- This lab was another way to gain experience within cadence virtuoso. We were tasked with creating a layout of the previous three input schematics of a NOR and NAND gate. This lab emphasizes the importance of building on our layout knowledge and getting hands on tinkering with the cadence virtuoso tools for creating a layout like we have done in class and talked about as a class. The use of metals, poly, and vias to make the layout correctly interact with each nmos and pmos transistor. The reinforcement of creating series and parallel connections within the layout and how it is done within the software is a learning curve but a good tool to visualize the circuit in another way. The result is generating a waveform from this layout and comparing it to the schematic done waveform to verify it is correctly displaying the output based on inputs.